

# Mathematical and Computational Methods

## Exercise Sheet 8: Differentiation II: applications

This sheet accompanies *Unit 8 (Differentiation II: applications)*. It exercises the four core skills of the unit—tangents and normals, locating and classifying stationary points, curve sketching, and applied optimisation—together with the supporting theorems (intermediate-value, extreme-value, mean-value and inverse-function), l'Hôpital's rule and the idea of gradient descent. Everything here uses only the differentiation of Units 7–8 and the Unit 1 foundations; no material from the complex-number, linear-algebra or integration units is needed.

**How to use this sheet.** Problems are graded by difficulty—★ drill (routine fluency), ★★ core (standard, tutorial level) and ★★★ challenge (harder or extension)—and organised in three sections: **warm-up drills** for fluency, **tutorial problems** (the core set for the 2-hour class, with the live items flagged ►) and **further practice** for independent home study.

**Solutions.** This is the *student* sheet (problems only); a full worked solution sits after each problem but is hidden. To produce the solutions version, uncomment `\solutionson` in the preamble (or build with `pdflatex "\def\showssolutions{}\input{MCM_Exercises_Unit8}"`) and recompile.

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## 1 Warm-up drills

**Exercise 1.1** (★ drill, *tangents & normals*). For  $f(x) = x^2$ , find the equations of the tangent and the normal to the curve at the point  $(2, 4)$ .

**Exercise 1.2** (★ drill, *l'Hôpital*). Evaluate, stating that the form is indeterminate before applying the rule:

$$(a) \lim_{x \rightarrow 0} \frac{\sin x}{x}, \quad (b) \lim_{x \rightarrow 0} \frac{e^x - 1}{x}, \quad (c) \lim_{x \rightarrow \infty} \frac{\ln x}{x^2}.$$

**Exercise 1.3** (★ drill, *classify*). Locate and classify the stationary points of  $f(x) = x^3 - 3x^2$  using the second-derivative test.

**Exercise 1.4** (★ drill, *asymptotes*). Find the vertical and horizontal asymptotes of

$$(a) f(x) = \frac{1}{x-2}, \quad (b) g(x) = \frac{2x}{x-3}.$$

**Exercise 1.5** (*★ drill, optimisation*). Two non-negative numbers add to 10. What is the largest their product can be?

**Exercise 1.6** (*★ drill, gradient descent*). Apply gradient descent to  $f(x) = x^2$  with learning rate  $\eta = 0.1$ , starting from  $x_0 = 1$ . Compute  $x_1$  and  $x_2$ , and state the limit of the iteration.

**Exercise 1.7** (*★ drill, critical point where  $f'$  undefined*). Show that  $f(x) = x^{2/3}$  has a critical point at  $x = 0$  even though  $f'(0)$  does not exist, and decide whether it is a local maximum or minimum.

**Exercise 1.8** (*★ drill, intermediate value theorem*). Show that  $x^3 + x - 1 = 0$  has a solution in the interval  $(0, 1)$ .

**Exercise 1.9** (*★ drill, inverse-function derivative*). Let  $f(x) = x^5 + x + 1$ . Given that  $f$  is one-to-one, find  $(f^{-1})'(1)$  without finding  $f^{-1}$  explicitly.

**Exercise 1.10** (*★ drill, sketch from  $f'$* ). Sketch the parabola  $f(x) = x^2 - 4x + 3$  by first using its derivative: find the interval(s) of increase and decrease and the vertex, then add the intercepts.

**Exercise 1.11** (*★ drill, mean value theorem*). Find the value of  $c$  guaranteed by the mean value theorem for  $f(x) = x^2$  on  $[1, 3]$ .

**Exercise 1.12** (*★ drill, extreme-value / closed interval*). Use the closed-interval (extreme-value-theorem) method to find the absolute maximum and minimum of  $f(x) = x^2 - 4x + 1$  on  $[0, 3]$ .

## 2 Tutorial problems

The problems flagged ► form the ★★ core set worked live in the 2-hour tutorial; the remaining problems are for completion at home.

**Exercise 2.1** (► ★★ core, tangents & normals). Consider the curve  $y = x^3 - 2x + 1$ .

1. Find the tangent and normal lines at the point where  $x = 1$ .
2. Find the equations of *all* tangent lines to the curve that are parallel to the line  $y = x$ .

**Exercise 2.2** (► ★★ core, l'Hôpital — several forms). Evaluate, first rewriting each into a  $\frac{0}{0}$  or  $\frac{\infty}{\infty}$  form where necessary:

$$(a) \lim_{x \rightarrow 0} \frac{x - \sin x}{x^3}, \quad (b) \lim_{x \rightarrow 0^+} x \ln x, \quad (c) \lim_{x \rightarrow 0} \left( \frac{1}{x} - \frac{1}{\sin x} \right).$$

**Exercise 2.3** (► ★★ core, classify — worked-example anchor). Locate and classify all stationary points of

$$f(x) = 2x^3 - 3x^2 - 12x + 5.$$

**Exercise 2.4** (► ★★ core, full polynomial sketch). Carry out a full curve-sketching analysis of  $f(x) = x^3 - 6x^2 + 9x$  (domain/symmetry, intercepts, increase/decrease and extrema, concavity and inflection) and sketch the curve.

**Exercise 2.5** (★★ core, full rational sketch). Sketch  $f(x) = \frac{x^2 + 1}{x^2 - 1}$ : state the domain, symmetry, intercepts, asymptotes, stationary points and concavity, then draw the curve.

**Exercise 2.6** (► \*\*core, optimisation — worked-example anchor). An open-topped box is made from a square sheet of card of side 12 cm by cutting a small square of side  $x$  from each corner and folding up the flaps. Find the value of  $x$  that maximises the volume, and the maximum volume.

**Exercise 2.7** (\*\*core, physics: maximum power transfer). A battery of EMF  $\varepsilon$  and internal resistance  $r$  delivers power  $P(R) = \frac{\varepsilon^2 R}{(R+r)^2}$  to an external load of resistance  $R > 0$ . Show that the power delivered is greatest when  $R = r$ , and find that maximum power.

**Exercise 2.8** (\*\*core, physics: projectile range). A projectile is launched from level ground with fixed speed  $v$  at an angle  $\theta$  to the horizontal. Neglecting air resistance, its range is

$$R(\theta) = \frac{v^2}{g} \sin(2\theta), \quad 0 \leq \theta \leq \frac{\pi}{2}.$$

Find the launch angle that maximises the range, and the maximum range.

**Exercise 2.9** (\*\*core, data science: least squares and the mean). Given data values  $x_1, \dots, x_n$ , the sum of squared errors of a single predicted constant  $c$  is  $L(c) = \sum_{i=1}^n (x_i - c)^2$ .

1. Show that  $L$  is minimised when  $c$  equals the mean  $\bar{x} = \frac{1}{n} \sum_i x_i$ .
2. Find the minimising  $c$  for the data  $\{2, 4, 9\}$ .

**Exercise 2.10** (► \*\*core, gradient descent on a quadratic). Minimise  $f(x) = x^2 - 4x + 5$  by gradient descent with learning rate  $\eta = 0.25$ , starting from  $x_0 = 0$ .

1. Write the update rule  $x_{n+1} = g(x_n)$  explicitly and compute  $x_1, x_2, x_3$ .
2. To what value do the iterates converge, and why is that the right answer?
3. What would go wrong if the learning rate were taken as  $\eta = 1.5$ ?

**Exercise 2.11** (\*\*core, mean value theorem). 1. Find the value of  $c$  guaranteed by the mean value theorem for  $f(x) = x^3$  on  $[0, 2]$ .

2. Use the mean value theorem to prove that  $|\sin a - \sin b| \leq |a - b|$  for all real  $a, b$ .

**Exercise 2.12** (\*\*core, inverse-function theorem). Let  $f(x) = 2x + \cos x$ .

1. Show that  $f$  is strictly increasing on  $\mathbb{R}$ , hence one-to-one and invertible.
2. Compute  $(f^{-1})'(1)$ .

**Exercise 2.13** (► \*\*core, spot the error (conceptual)). To find the largest area of a rectangle with perimeter 20, a student sets width  $x$ , height  $10 - x$ , writes  $A(x) = x(10 - x)$ , finds  $A'(x) = 10 - 2x = 0$ , so  $x = 5$ , and then concludes “ $A''(x) = -2 < 0$ , so this is a minimum.” Identify the error and state the correct conclusion.

**Exercise 2.14** (\*\*core, Newton’s method). The equation  $x^3 - 2x - 5 = 0$  has a single real root, which the intermediate value theorem traps in  $(2, 3)$  (since  $f(2) = -1 < 0$  and  $f(3) = 16 > 0$ ). Use Newton’s method to pin it down: starting from  $x_0 = 2$ , write the iteration explicitly and carry out two steps by hand, tabulating  $x_n$ ,  $f(x_n)$  and  $f'(x_n)$  at each stage to obtain  $x_1$  and  $x_2$ . Comment on how fast the estimate is settling.

### 3 Further practice (home)

#### Tangents and normals

**Exercise 3.1** (*★★ core, tangent & normal*). For the curve  $y = \frac{1}{x}$ , find the tangent and normal at the point  $(2, \frac{1}{2})$ , and find where the normal crosses the  $x$ -axis.

**Exercise 3.2** (*★★ core, tangents from an external point*). Find the equations of the two tangent lines to the parabola  $y = x^2$  that pass through the point  $(0, -1)$ .

**Exercise 3.3** (*★★★ challenge, challenge: a chord property*). Show that the tangent to  $y = x^3$  at the point  $(a, a^3)$  (with  $a \neq 0$ ) meets the curve again at exactly one further point, and that its  $x$ -coordinate is  $-2a$ .

#### Stationary points and classification

**Exercise 3.4** (*★★ core, classify a mixed bag*). Locate and classify all stationary points of:

$$(a) f(x) = x^3 - 3x^2 + 3x, \quad (b) g(x) = x + \frac{1}{x}, \quad (c) h(x) = xe^{-x}.$$

**Exercise 3.5** (*★ drill, when the second-derivative test fails*). For  $f(x) = x^4$ , show that  $f''(0) = 0$ , so the second-derivative test is inconclusive, and then classify the stationary point at  $x = 0$  by another method.

**Exercise 3.6** (*★★ core, absolute extrema on a closed interval*). Use the closed-interval (extreme-value-theorem) method to find the absolute maximum and minimum of:

$$(a) f(x) = x^3 - 3x \text{ on } [0, 3], \quad (b) g(x) = x^4 - 8x^2 + 3 \text{ on } [-1, 3].$$

#### Curve sketching

**Exercise 3.7** (*★★ core, polynomial sketch*). Sketch  $f(x) = x^3 - 12x$ : give symmetry, intercepts, stationary points and their type, and the concavity.

**Exercise 3.8** (*★★ core, rational sketch*). Sketch  $f(x) = \frac{x}{x^2 - 1}$ : domain, symmetry, intercepts, asymptotes, monotonicity.

**Exercise 3.9** (*★★ core, rational sketch*). Sketch  $f(x) = \frac{x-1}{x+2}$ , identifying the asymptotes and the intercepts.

**Exercise 3.10** (*★★ core, sketch from features*). Sketch  $f(x) = x^4 - 8x^2 + 16$ . (Hint: it factorises.)

#### Optimisation

**Exercise 3.11** (*★★ core, inscribed rectangle*). A rectangle has its base on the  $x$ -axis and its two upper corners on the parabola  $y = 12 - x^2$  (with  $y \geq 0$ ). Find the dimensions of the rectangle of greatest area, and that area.

**Exercise 3.12** (*★★ core, divided pen*). A rectangular animal pen of area  $600 \text{ m}^2$  is divided into two equal halves by a fence parallel to one pair of sides. Find the dimensions that minimise the total length of fencing used (perimeter plus the internal divider).

**Exercise 3.13** (★★ core, cylindrical can). A closed cylindrical can must hold a volume of  $16\pi \text{ cm}^3$ . Find the radius and height that minimise its total surface area, and verify that at the optimum the height equals the diameter.

**Exercise 3.14** (★★ core, data science: closest point). Find the point(s) on the parabola  $y = x^2$  closest to  $(0, \frac{3}{2})$ .

**Exercise 3.15** (★★ core, physics: kinematics min/max). A particle moves along a straight line with position  $s(t) = t^3 - 9t^2 + 24t$  metres at time  $t$  seconds, for  $0 \leq t \leq 5$ . Find (a) the times at which it is instantaneously at rest, and (b) its maximum and minimum velocity on the interval.

**Exercise 3.16** (★★★ challenge, challenge: the Norman window). A “Norman” window is a rectangle topped by a semicircle, the semicircle’s diameter being the top of the rectangle. For a fixed perimeter  $P$ , find the proportions that maximise the area, and show that at the optimum the rectangle’s height equals the radius of the semicircle.

## Limits (l’Hôpital)

**Exercise 3.17** (★★ core, a batch of limits). Evaluate:

$$(a) \lim_{x \rightarrow 0} \frac{\tan x - x}{x^3}, \quad (b) \lim_{x \rightarrow 1} \frac{\ln x}{x - 1}, \quad (c) \lim_{x \rightarrow \infty} \frac{x^2}{e^x}, \quad (d) \lim_{x \rightarrow 0^+} x^x.$$

**Exercise 3.18** (★★ core, spot the error). A student writes

$$\lim_{x \rightarrow 0} \frac{x^2 + 1}{x + 1} \stackrel{H}{=} \lim_{x \rightarrow 0} \frac{2x}{1} = 0.$$

Explain what is wrong, and give the correct value.

**Exercise 3.19** (★★★ challenge, challenge: a  $1^\infty$  limit). Show that  $\lim_{x \rightarrow \infty} \left(1 + \frac{a}{x}\right)^x = e^a$  for a constant  $a$ .

## Theorems: IVT, MVT, inverse functions

**Exercise 3.20** (★★ core, IVT). Show that the equation  $\cos x = x$  has at least one solution in  $(0, 1)$ .

**Exercise 3.21** (★★ core, existence and uniqueness). Show that  $x^3 + x - 1 = 0$  has *exactly* one real root. (Hint: IVT for existence, the sign of the derivative for uniqueness.)

**Exercise 3.22** (★★ core, MVT inequality). Use the mean value theorem (or the sign of a derivative) to prove that  $e^x \geq 1 + x$  for all  $x$ , with equality only at  $x = 0$ .

**Exercise 3.23** (★★ core, inverse-function derivatives). 1. For  $f(x) = x^3 + 2x + 1$  (one-to-one), find  $(f^{-1})'(1)$ .

2. Use the inverse-function theorem to recover  $\frac{d}{dx} \arctan x = \frac{1}{1 + x^2}$  from  $\tan$ .

## Gradient descent

**Exercise 3.24** (★★ core, descent with a shifted minimum). Apply gradient descent to  $f(x) = (x - 3)^2$  with  $\eta = 0.1$ , starting from  $x_0 = 0$ . Write the update rule, compute  $x_1, x_2, x_3$ , and identify the limit.

**Exercise 3.25** (★★★ challenge, challenge: convergence condition). For  $f(x) = x^2$  the gradient-descent update with learning rate  $\eta$  is  $x_{n+1} = x_n - \eta (2x_n)$ .

1. Show that  $x_n = (1 - 2\eta)^n x_0$ .
2. Deduce the exact range of  $\eta > 0$  for which  $x_n \rightarrow 0$  for every starting point, and describe the behaviour when  $\eta = 1$  and when  $\eta > 1$ .

### Conceptual / true–false

**Exercise 3.26** (*★ drill, true or false*). For each statement, say whether it is true or false and justify briefly.

1. If  $f'(c) = 0$  then  $f$  has a local maximum or minimum at  $c$ .
2. If  $f''(c) = 0$  then  $c$  is an inflection point of  $f$ .
3. A continuous function on an open interval need not attain a maximum value.
4. If  $f'(x) > 0$  on an interval, then  $f$  is strictly increasing there.
5. At a point where  $f'(a) = 0$ , the normal line is horizontal.
6. l'Hôpital's rule can be applied to any limit of a quotient.

**Exercise 3.27** (*★★ core, interpret a graph of  $f'$* ). The graph below shows the *derivative*  $f'$  of a function  $f$  (not  $f$  itself). Using only this graph, describe  $f$ : state the intervals on which  $f$  is increasing or decreasing, locate and classify its stationary points, and say where  $f$  is concave up or down and where it has a point of inflection.

